

Q1]

a) Simple returns are given by:

$$R(t) = \frac{(P_t - P_{t-1})}{P_{t-1}}$$

- 1st Month: $R(1) = \frac{108 - 100}{100} = 8\%$
- 2nd Month: $R(2) = \frac{103 - 108}{108} = -4.63\%$
- 3rd Month: $R(3) = \frac{115 - 103}{103} = 11.65\%$
- 4th Month: $R(4) = \frac{110 - 115}{115} = -4.35\%$

b) Log returns are given by:

$$r(t) = \ln \left[\frac{P_t}{P_{t-1}} \right]$$

- 1st Month: $r(1) = \ln \left(\frac{108}{100} \right) = 7.70\%$
- 2nd Month: $r(2) = \ln \left(\frac{103}{108} \right) = -4.73\%$
- 3rd month: $r(3) = \ln \left(\frac{115}{103} \right) = 11\%$
- 4th month: $r(4) = \ln \left(\frac{110}{115} \right) = -4.44\%$

c) Deviations: $|R_t - r(t)| = D(t)$

$$D(1) = 0.3\%$$

$$D(2) = 0.1\%$$

$$D(3) = 0.65\% \rightarrow \text{Least accurate.}$$

$$D(4) = 0.09\%$$



$$\text{Reason: } R(t) \approx r(t)$$



Valid only for small returns.



Larger returns $\Rightarrow$ Larger deviations.

Q2]

a) Linearity of Expectations: $E(R_p)$

$\Rightarrow$ For random variables X & Y & constants a & b , the Expectation value of $aX + bY$ is given as.

$$E(aX + bY) = aE(X) + bE(Y)$$



In portfolio returns, $R_p = w_1 R_1 + w_2 R_2$

$$\therefore E(w_1 R_1 + w_2 R_2) = E(R_p) = w_1 E(R_1) + w_2 E(R_2)$$

$$\therefore E(R_p) = 0.15(0.4) + (0.07)(0.6)$$

$$E(R_p) = 0.102 = 10.2\%$$

b) NO. The linearity of Expectations is independent of correlation b/w r.v.s.

$$23] \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \quad \text{--- (I)}$$

$$\text{Var}(aX) = a^2 \text{Var}(X) \quad \text{--- (II)}$$

$$R_p = w_1 R_1 + w_2 R_2$$

$$\text{Let } X = w_1 R_1 \text{ \& } Y = w_2 R_2$$

$$\left. \begin{aligned} \therefore \text{Var } X &= w_1^2 \text{Var } R_1 \\ \therefore \text{Var } Y &= w_2^2 \text{Var } R_2 \end{aligned} \right\} \text{From (II)}$$

$$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y) \quad \text{--- (III)}$$

↓

$$\text{Cov}(w_1 R_1, w_2 R_2) = w_1 w_2 \text{Cov}(R_1, R_2)$$

Putting all values in (I).

$$\left[\begin{aligned} \text{where, } \sigma_p^2 &= \text{Var}(w_1 R_1 + w_2 R_2) \\ \sigma_1^2 &= \text{Var}(R_1) \text{ \& } \sigma_2^2 = \text{Var}(R_2) \end{aligned} \right]$$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

By correlation:

$$\rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2}$$

↓

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Hence proved.

Q5] $\sigma_A = 25\%$, $\sigma_B = 12\%$, $w_A = w_B = 0.5$

We know that $\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$

• for $\rho = +1$,

$$\begin{aligned}\sigma_P^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B \\ &= 156.25 + 36 + 150 \\ &= 342.25\end{aligned}$$

$$\Downarrow$$

$$\sigma_P = \sqrt{342.25} = 18.5\%$$

• for $\rho = 0$

$$\sigma_P^2 = 156.25 + 36 = 192.25$$

$$\sigma_P = 13.87\%$$

• for $\rho = -1$

$$\sigma_P^2 = 156.25 + 36 - 150 = 42.25$$

$$\sigma_P = 6.5\%$$

b) Wt. avg of individual risk: $w_A \sigma_A + w_B \sigma_B = 0.5(25) + 0.5(12)$
 $= 18.5\%$

$$\begin{aligned}\sigma_P^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B \\ &= (\sigma_A w_A + \sigma_B w_B)^2\end{aligned}$$

OR.

$$\sigma_P = \sigma_A w_A + \sigma_B w_B = \text{Wt. avg. of individual risk}$$

Hence proved.

Signal to $\rho = +1$

c) For $\rho = -1$

$$\sigma_p^2 = (w_A \sigma_A - w_B \sigma_B)^2$$

$$\sigma_p = |w_A \sigma_A - w_B \sigma_B| \rightarrow 0$$

$$\therefore \frac{w_A}{w_B} = \frac{\sigma_B}{\sigma_A} = \frac{w_A}{1-w_A}$$

$$\therefore \boxed{(w_A + w_B) = 1}$$

$$\therefore 25 w_A = 12(1 - w_A)$$

$$37 w_A = 12$$

$$w_A = 32.43\%$$

$$w_B = 67.57\%$$

5] Weighted ^{avg} risk sq: $(w_1 \sigma_1 + w_2 \sigma_2)^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2$

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Portfolio variance: $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

Now, ρ_{12} is strictly less than 1

$$\text{i.e. } \rho_{12} < 1$$

$$\therefore \sigma_p^2 < (w_1 \sigma_1 + w_2 \sigma_2)^2$$

$$\therefore \sigma_p < (w_1 \sigma_1 + w_2 \sigma_2)$$

- The more we diversify, the less is the risk of loss by loss in a single asset as the other asset may cover up for it.

Q6]

a) $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$

$$\Sigma = \begin{bmatrix} 1\sigma_1^2 & (0.4)\sigma_1\sigma_2 & (0.1)(\sigma_1\sigma_3) \\ (0.4)\sigma_1\sigma_2 & 1\sigma_2^2 & (0.2)\sigma_2\sigma_3 \\ (0.1)\sigma_1\sigma_3 & (0.2)\sigma_2\sigma_3 & 1\sigma_3^2 \end{bmatrix} \quad \{ \rho_{ij} = \rho_{ji} \}$$

$$\Sigma = \begin{bmatrix} 625 & 120 & 10 \\ 120 & 144 & 9.6 \\ 10 & 9.6 & 16 \end{bmatrix}$$

b) In the upper matrix Σ

$\therefore \Sigma_{ij} = \Sigma_{ji}$ } the matrix is symmetrical.

Q7] Sharp Ratio: $S = \frac{E[R_p] - R_f}{\sigma_p}$

a)

• X: $S_x = \frac{14-4}{22} = 0.4545$

• Y: $S_y = \frac{9-4}{10} = \frac{5}{10} = 0.50$

• Z: $S_z = \frac{20-4}{35} = 0.4571$

b) $\therefore S(y) > S(z) > S(x)$,

Y earns 0.50% of excess return for every 1% of risk taken $\Rightarrow$ Rational Investor will prefer Portfolio Y.

c) Z has highest absolute return $\Rightarrow 20\%$

But

Z also has the highest extra risk $\Rightarrow 35\%$.

also $S_Z < S_Y$



Z gives less reward per unit risk.



A Rational investor might still prefer Y.

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a) The GMV portfolio is the portfolio with lowest possible risk among all wt. weight combinations.

• It is the leftmost pt. because no. portfolio can have lower σ than this.



it is the leftmost pt. on horizontal axis.

b) NO.

Every portfolio on the lower half has an identical portfolio on the upper half with the same risk but higher return.



Risk tolerance ^{may} ~~can~~ decide where you choose to be on the upper half $\Rightarrow$ NOT on the lower half

- c) CML is constructed by following steps.
- Plotting Efficient frontier.
 - Mark risk free asset at pt $(0, R_f)$.
 - Draw line from R_f tangent to efficient frontier $\downarrow$ CML.

$$E(R_p) = R_f + \frac{E(R_M) - R_f}{\sigma_M} \cdot \sigma_p$$

Slope $\Rightarrow$ sharp ratio

M = max. risk adjusted return.

Q9]

- a) $n = 36$, $\sigma_i = 30\%$.

$\therefore$ Std error of sample mean

$$SE(\hat{\mu}_1) = \frac{\sigma_i}{\sqrt{n}} = \frac{30\%}{6} = 5\%.$$

$\downarrow$

Implication $\Rightarrow$ We cannot rely only on historical data alone while building portfolios. as the expected return may have error.

- b) MVO takes μ & Σ as inputs & finds the optimal wts. But Optimiser treats estimator μ to be accurate so if $\hat{\mu}$ is slightly wrong, ~~opt~~ error is amplified into the portfolio ~~wts~~ weights.
- Replacing $\hat{\mu}$ with ML predicted returns helps because,
 - ML models see patterns beyond historical avg.
 - They produce more accurate & stable return estimates.
- c) $N = 50$, Σ is sym $\Rightarrow$ no. of unique $\underbrace{\text{entities}}_2 = \frac{N(N+1)}{2} = 1275$ parameters.
- Estimation of 1275 parameters is difficult & ~~more~~ ^{more} ~~as~~ ^{as} we add the more ~~becomes~~ ^{becomes} to predict.